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Biodiversity risk and global stock markets: A cross-national heterogeneity analysis based on quantile-on-quantile methods

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ABSTRACT

This paper uses quantile-on-quantile kernel-regularised least squares (QQKRLS) and quantile-on-quantile Granger causality (QQGC) methods to examine how the S&P Global Large MidCap Biodiversity Index (GBI) heterogeneously influences the stock markets of the G7 countries and China. Our empirical analysis of data from 2019 to 2025 demonstrates that the risk of biodiversity loss has a significant, quantile-dependent, and non-linear effect on these national stock markets. The findings reveal that the stock markets of developed European economies display highly sensitive positive responses to GBI fluctuations, whilst the United States' stock market exhibits complex cross-causality structures. Conversely, the Chinese stock market shows pronounced asymmetric and stage-specific characteristics, demonstrating significant vulnerability during periods of heightened risk. Robustness tests using quantile regression and ordinary least squares (OLS) regression further validate the reliability of these principal findings. This research provides substantial empirical evidence on the cross-national transmission mechanisms of sustainable finance. It holds significant theoretical value and offers practical implications for investors developing quantile-sensitive investment strategies and for policymakers refining green finance regulatory frameworks. The results forecast that biodiversity-related financial risks will continue to have heterogeneous effects across different market conditions and geographical regions.

1. Introduction

Global climate change and environmental degradation have intensified, positioning biodiversity conservation as a paramount challenge confronting the international community (Baste and Watson, 2022; Farooqi et al., 2022; Rands et al., 2010; Zeng et al., 2025). The accelerating rate of biodiversity loss, rather than conservation per se, constitutes the primary threat to global ecological stability and financial systems. Recent advances in international environmental agreements, including the United Nations Biodiversity Framework and the Paris Agreement, have transformed biodiversity conservation from a core environmental policy issue into an influential factor shaping global financial markets (McElwee et al., 2020; Warren et al., 2018). Biodiversity-specific financial instruments focus exclusively on ecosystem and species preservation metrics, distinguishing themselves from broader Environmental, Social, and Governance (ESG) investment

products that encompass multiple sustainability dimensions. Biodiversity assets, representing a novel category of sustainable investment instruments, have fundamentally reshaped traditional investment concepts and asset allocation strategies (Becchetti et al., 2022; Gareil et al., 2024; Karolyi and Tobin-de la Puente, 2023).

The launch of the S&P Global LargeMidCap Biodiversity Index (GBI) has marked biodiversity investment's entry into standardised and large-scale development. This index measures the performance of a screened and weighted subset of stocks from the S&P Global LargeMidCap Index, employing selection criteria that reduce ecosystem impact intensity and enhance alignment with specific United Nations Sustainable Development Goals. This innovative index provides investors with a benchmark tool for quantifying biodiversity investment performance whilst offering academia a unique observational perspective for studying biodiversity factors' influence on financial market performance.

Environmental finance literature has established a rich theoretical

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foundation supported by substantial empirical evidence. The Porter Hypothesis (Porter and Linde, 1995) and subsequent research have progressively expanded to examine ESG investment's impact on stock returns and risk. Kräussl et al. (2024) identified a significant positive correlation between ESG performance and financial performance through comprehensive literature review, although multiple factors moderate this relationship. Recent studies have focused on climate risk and green finance impacts on capital markets, with Garel et al. (2024) demonstrating that investors have begun pricing biodiversity loss risk—defined as the financial implications of ecosystem degradation and species extinction rather than conservation efforts—particularly following the Kunming Declaration's release.

However, existing research exhibits significant gaps regarding biodiversity-financial market relationships. Most research has concentrated on climate change and carbon emissions whilst inadequately addressing biodiversity as a distinct category separate from general ESG considerations. Flammer et al. (2025) identified biodiversity finance as an emerging field requiring greater academic attention. Existing literature predominantly employs traditional linear regression methods; Garel et al. (2024) utilised standard ordinary least squares and fixed-effects models, whilst Hutchinson and Lucey (2024) primarily relied on bibliometric linear correlation analysis. These conventional approaches, though valuable, possess inherent limitations in capturing complex nonlinear, asymmetric, and quantile-dependent relationships that characterise environmental-financial market interactions (Shahbaz et al., 2017; Sim and Zhou, 2015). Research has predominantly concentrated on single or developed markets, lacking systematic comparative analysis across different development levels. Furthermore, existing methodologies rely primarily on traditional cointegration analysis. The Engle and Granger (1987) cointegration tests and Johansen (1991) procedures assume constant parameters across entire distributions, thereby ignoring heterogeneity at different quantile levels. Studies employing these methods in environmental finance cannot distinguish whether biodiversity impacts differ between bull and bear markets or whether risk transmission intensifies during financial crises (Xiao, 2009). The quantile-on-quantile approach addresses these limitations by allowing parameters to vary across both dependent and independent variables' distributions, revealing state-dependent risk transmission mechanisms that remain hidden in mean-based analyses (Sim and Zhou, 2015).

Linear models' methodological constraints become particularly evident when examining tail dependencies and extreme market conditions. Recent environmental finance studies have begun recognising these limitations. Traditional cointegration and vector autoregression models can identify average relationships but fail to reveal how biodiversity risks transmit differently during market stress versus normal periods (Balcilar et al., 2017). This methodological gap proves critical because environmental risks often manifest non-linearly, with threshold effects and regime switches that linear models cannot adequately capture (Baur, 2013).

Hutchinson and Lucey (2024) found through systematic literature review that biodiversity finance research has concentrated mainly on theoretical studies with limited empirical investigation. Chen et al. (2023) and Ma et al. (2025) demonstrated that ESG ratings improve corporate financial performance. However, these broad ESG metrics may obscure biodiversity factors' specific financial impacts, which require dedicated analysis through specialised biodiversity-focused indices such as the GBI. Narula et al. (2025) demonstrated that ESG investment performance responds to global sustainability attention, suggesting time-varying and complex ESG investment effects that require more refined analytical methods.

Based on these research gaps, this study poses the following core research questions: Do biodiversity assets significantly impact stock markets in G7 countries and China? Do these impacts exhibit heterogeneity under different market conditions and quantile levels? Do fundamental differences exist in impact mechanisms between developed

and emerging markets? To systematically address these questions, this study employs the S&P Global LargeMidCap Biodiversity Index as a proxy variable for biodiversity assets and selects major stock market indices from G7 countries and China as research subjects.

Regarding methodological innovation, this study employs two cutting-edge quantile regression techniques: Quantile-on-Quantile Kernel-Based Regularised Least Squares (QQKRLS) and Quantile-on-Quantile Granger Causality (QQGC). These methods simultaneously consider quantile information from both dependent and independent variables, revealing differentiated impact patterns under different market state combinations and identifying threshold effects, asymmetric effects, and state-dependent effects that traditional linear models cannot capture.

From a policy perspective, biodiversity loss has emerged as an important threat to global financial stability. Mies (2025) found that sovereign biodiversity risk drivers significantly impact financial stability. Major central banks have begun recognising this systemic risk: the Bank of England's 2021 climate-related financial risk management expectations now extend to nature-related risks (Bank of England, 2021), whilst the Network for Greening the Financial System (NGFS), comprising 127 central banks and supervisors, published its comprehensive nature-related financial risks framework in 2023, marking the formal integration of biodiversity considerations into global financial stability assessments (NGFS, 2023).

This work contributes across multiple dimensions: expanding environmental finance's theoretical boundaries, providing empirical support for biodiversity asset pricing theory, innovatively introducing advanced quantile regression methods into biodiversity-financial market relationship research, and providing the first systematic empirical evidence of biodiversity assets' impact on stock markets at different development levels. The research results provide an important empirical foundation for policymakers and scientific risk assessment tools for investors.

2. Methodology and data

Compared to classical econometric approaches, the QQKRLS method offers distinct methodological advantages. Traditional OLS regression captures only mean effects, missing tail behaviours where biodiversity risks often manifest most severely (Koenker and Bassett, 1978). Standard quantile regression, whilst addressing distributional heterogeneity, assumes the independent variable affects all quantiles uniformly (Koenker, 2005). Non-linear models such as threshold or Markov-switching approaches require pre-specification of functional forms or regime numbers, introducing model uncertainty (Hamilton, 1989; Hansen, 1999). In contrast, QQKRLS simultaneously examines how different quantiles of biodiversity risk (e.g., normal versus crisis states) affect various quantiles of stock returns (e.g., bear versus bull markets), revealing state-dependent relationships without imposing restrictive parametric assumptions.

Furthermore, the QQKRLS method demonstrates its ability to automatically identify and model complex nonlinear relationships between variables through the introduction of kernel function techniques, without requiring prior specification of specific functional forms—a critical advantage over polynomial or spline-based non-linear models that risk misspecification bias (Stone, 1985). Meanwhile, the application of regularisation techniques effectively avoids overfitting problems, significantly improving the out-of-sample forecasting capability and statistical robustness of estimation results. More importantly, the QQKRLS method can simultaneously consider quantile information from both dependent and independent variables, revealing differentiated impact patterns under different market state combinations, which represents an important function that traditional methods cannot achieve. The QQGC method, building upon traditional Granger causality tests, introduces a quantile-on-quantile analysis framework, enabling examination of dynamic causal relationships between biodiversity assets and stock markets at different quantile levels, providing more precise and

comprehensive analytical tools for understanding the complex transmission mechanisms between the two. This methodological innovation enables us to identify threshold effects, asymmetric effects, and state-dependent effects that traditional linear models cannot capture.

2.1. Quantile-on-quantile kernel-based regularised least squares regression (QKRLS)

This research employs the QKRLS methodology, as proposed by Adebayo et al. (2024), to assess how biodiversity assets influence G7 and Chinese stock markets. The QKRLS approach offers distinct methodological strengths in examining dynamic relationships and interdependencies among economic variables. Its primary advantage lies in its capacity to capture heterogeneous and non-linear relationships across different quantile distributions, providing insights that conventional linear modeling approaches cannot adequately address. This feature proves particularly beneficial when investigating how predictor variables exert varying effects across the entire data distribution, thereby enhancing our comprehension of complex economic relationships. The QKRLS framework for analyzing the influence of an explanatory variable on a response variable is specified as:

$$E_N \left[\frac{dQY_\tau}{dQX_{\theta k}} \right] = \frac{-2}{\sigma^2 N} \sum_k \sum_i j_i e^{\frac{\|X_{\theta i} - X_{\theta k}\|^2}{\sigma^2}} (X_{\theta i} - X_{\theta k}) \quad (1)$$

Here, $E_N[\beta QY_\tau]$ represents the average pointwise marginal impact of the τ -th conditional quantile of the explanatory variable on the θ -th conditional quantile of the response variable. Nevertheless, the QKRLS approach possesses certain limitations. The primary constraint of this methodology is its restriction to empirically validated linear frameworks. Initially, the specific distributional assumptions underlying certain components may constrain the generalizability of results, particularly when datasets contain extreme values or exhibit irregular patterns that could affect the robustness of the analysis. Additionally, the computational requirements can be substantial, demanding intensive calculations and sophisticated analytical expertise for proper interpretation. Furthermore, while the method delivers detailed and precise quantile-specific effect estimates, translating these granular findings into actionable policy recommendations and broader economic development insights can prove challenging, potentially limiting their practical applicability in real-world decision-making contexts.

2.2. Quantile-to-quantile Granger causality (QQGC)

The conventional Granger causality (GC) framework, as established by Granger (1969), determines whether variable X possesses predictive power for variable Y . Traditional Granger causality tests have been widely applied in financial spillover studies but suffer from critical limitations. Baur and Schulze (2005) demonstrated that standard Granger tests fail to detect causality that exists only in extreme market conditions. Similarly, Chuang et al. (2009) showed that causality patterns can reverse across different quantiles—a phenomenon entirely

missed by conventional tests that focus solely on conditional means. This relationship is expressed as follows:

$$Y_t = \gamma_0 + \sum_{i=1}^p \gamma_i Y_{t-i} + \sum_{i=1}^p \beta_i X_{t-i} + \epsilon_t \quad (2)$$

In this equation, γ_0 , t , and p denote the constant term, time period, and lag length, respectively. Additionally, ϵ_t represents the stochastic error component. The parameters β_i and γ_i capture the influence of historical values of Y and X , correspondingly. While the GC methodology is robust, it fails to account for non-linear causal relationships among variables. To overcome these constraints, Adebayo and Özkan (2024) introduced the QQGC framework, which investigates how quantile distributions of one variable influence the quantile distributions of another. The QQGC specification is structured as:

$$QY_{(\tau,t)} = \gamma_{(\tau,\theta,0)} + \sum_{i=1}^p \gamma_i QY_{(\tau,t-i)} + \sum_{i=1}^p \beta_i QX_{(\theta,t-i)} + \epsilon_{(\tau,\theta,t)} \quad (3)$$

In addition, the intercept term that is specific to the quantiles τ and θ is represented by $\gamma_{(\tau,\theta,0)}$. The coefficients γ_i measure the influence of past values of Y at quantile τ on $QY_{(\tau,t)}$. The coefficients β_i measure the influence of past values of X at quantile θ on $QY_{(\tau,t)}$. Meanwhile, $\epsilon_{(\tau,\theta,t)}$ represents the error term for the model time at t that affects the quantiles t and θ .

Furthermore, $\gamma_{(\tau,\theta,0)}$ represents the quantile-specific intercept parameter corresponding to quantiles θ and τ . The coefficients $QY_{(\tau,t)} \cdot \beta_i$ capture the influence of historical Y values at quantile θ on the current quantile distribution, while $\gamma_{(\tau,\theta,0)} \cdot \gamma_i$ measure the effect of lagged X values at quantile τ on the dependent variable's quantile θ . The term $\epsilon_{(\tau,\theta,t)}$ signifies the stochastic disturbance component for the model at period t , which pertains to the interaction between quantiles θ and τ .

2.3. Data

This study employs a comprehensive dataset that encompasses major stock market indices from G7 countries and China, alongside a global biodiversity benchmark indicator. The stock market indices specifically include the FTSE MIB index (ITA) for Italy, the CAC index (FRA) for France, the FTSE 100 index (UK) for the United Kingdom, the DAX index (GER) for Germany, the S&P/TSX Composite index (CAN) for Canada, the Nikkei 225 index (JP) for Japan, the S&P 500 index (US) for the United States, and the Shanghai Composite index (CHN) for China. These indices represent the primary stock market performance and investor sentiment dynamics within their respective countries, with researchers sourcing all relevant data from the LSEG Data & Analytics database.

The S&P Global LargeMidCap Biodiversity Index (GBI) serves as the global biodiversity indicator, measuring the performance of a carefully selected and weighted subset of equity securities from the S&P Global LargeMidCap universe. Distinguished from conventional ESG indices

Table 1
Descriptive statistical results.

	Mean	Variance	Skewness	Kurtosis	JB	ERS
GBI	0.185	6.488	-0.857	5.929	485.618***	-8.458***
GER	0.153	8.69	-1.76	12.811	2250.625***	-8.457***
UK	0.058	5.456	-1.996	15.2	3148.888***	-8.263***
ITA	0.2	10.488	-2.256	16.197	3604.397***	-7.106***
CAN	0.154	5.402	-2.138	14.225	2813.045***	-6.920***
JP	0.188	8.781	-0.372	6.531	550.944***	-8.624***
US	0.231	7.52	-0.873	6.521	581.127***	-7.998***
CHN	0.038	4.896	0.48	3.19	141.529***	-6.575***
FRA	0.108	8.542	-1.742	12.341	2096.505***	-8.336***

Notes: JB refers to the Jarque-Bera test statistic for normality, ERS refers to the Elliott-Rothenberg-Stock unit root test for stationarity, and *** indicates statistical significance at the 1 % level.

Table 2

The Pearson correlation test.

	GBI	GER	UK	ITA	CAN	JP	US	CHN	FRA
GBI	1	0.577	0.481	0.51	0.603	0.477	0.833	0.247	0.56
GER	0.577	1	0.552	0.679	0.496	0.394	0.503	0.2	0.726
UK	0.481	0.552	1	0.54	0.499	0.295	0.425	0.134	0.588
ITA	0.51	0.679	0.54	1	0.477	0.368	0.44	0.187	0.696
CAN	0.603	0.496	0.499	0.477	1	0.351	0.574	0.16	0.487
JP	0.477	0.394	0.295	0.368	0.351	1	0.401	0.194	0.386
US	0.833	0.503	0.425	0.44	0.574	0.401	1	0.176	0.486
CHN	0.247	0.2	0.134	0.187	0.16	0.194	0.176	1	0.199
FRA	0.56	0.726	0.588	0.696	0.487	0.386	0.486	0.199	1

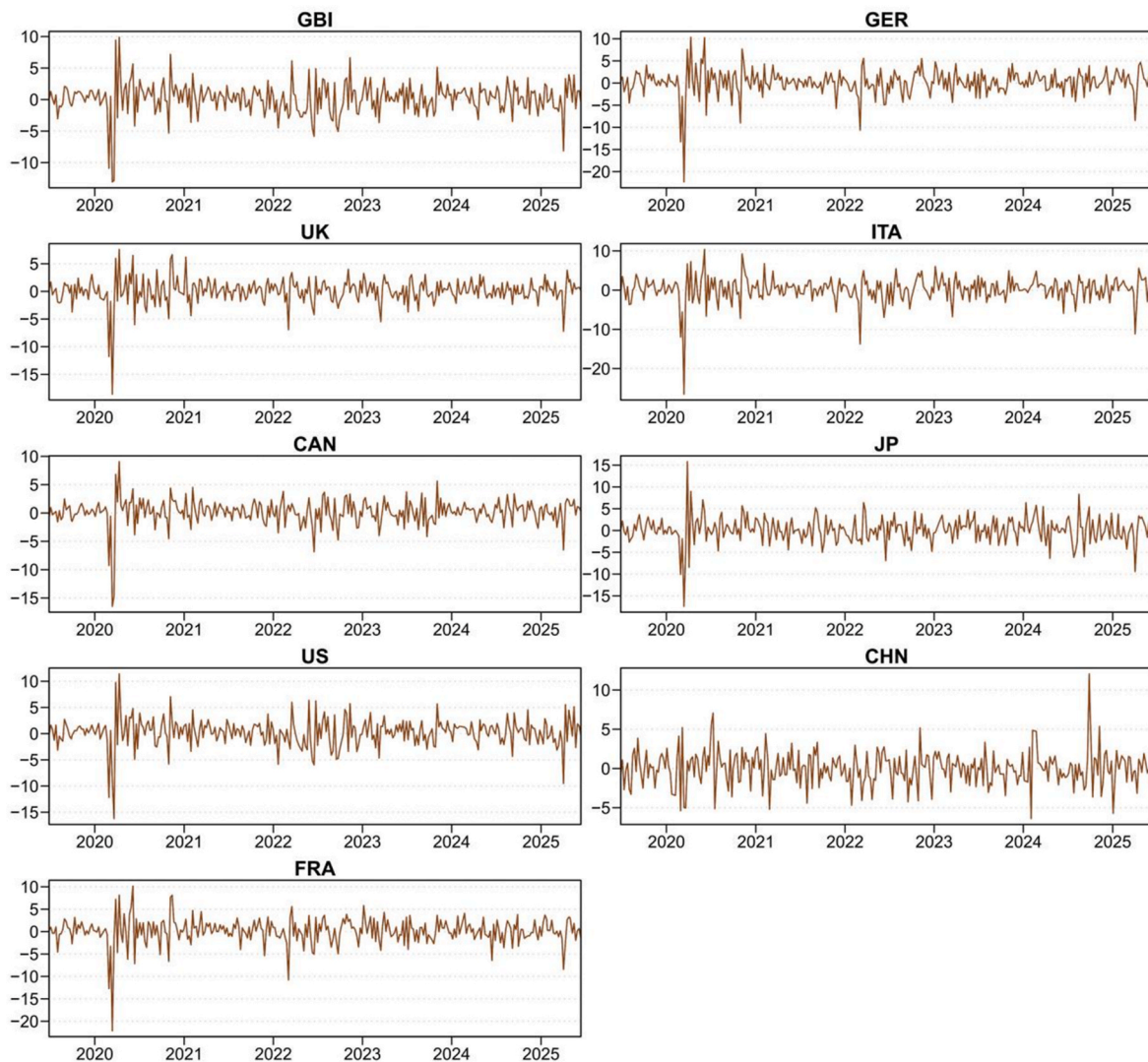
that weight environmental, social, and governance factors equally, this biodiversity-specific index concentrates solely on ecological impact metrics. The analysis systematically selects and optimises these assets to collectively achieve reduced ecosystem impact intensity, constrained proportional impact on significant terrestrial ecosystems, enhanced alignment with specific United Nations Sustainable Development Goals, and diminished carbon footprint relative to the S&P Global LargeMid-Cap benchmark. The study obtains the biodiversity index data from the official S&P Dow Jones Indices website <https://www.spglobal.com/spdji/en/indices/sustainability/sp-global-largemidcap-biodiversity-index/#overview>.

The research utilises data spanning from June 21, 2019 to June 13, 2025. To ensure data stationarity and eliminate the influence of price level differentials, all variables undergo logarithmic differencing transformation, thereby focusing the analytical framework on the dynamic relationships amongst return series.

3. Result analysis

3.1. Main empirical results

The descriptive statistical results presented in Table 1 reveal the

**Fig. 1.** The trends of all variables.

fundamental statistical characteristics and distributional properties of the GBI and various national stock markets. Regarding mean returns, the US stock market demonstrates the strongest performance (0.231), followed by Italy (0.2) and the GBI itself (0.185), whilst China exhibits the lowest mean return (0.038), reflecting substantial performance disparities across different markets. Variance analysis indicates that Italy displays the highest volatility (10.488), followed by Germany (8.69), whereas China demonstrates relatively lower volatility (4.896), suggesting significant differences in volatility characteristics between emerging and developed markets.

All variables exhibit negative skewness distributions in Table 1, with ITA (−2.256) and CAN (−2.138) demonstrating the most pronounced left-skewed characteristics, implying that extreme downward movements occur more frequently in these markets. Kurtosis statistics reveal that ITA (16.197) and the UK (15.2) possess the highest leptokurtic features, indicating a higher frequency of extreme values, whilst China displays relatively lower kurtosis (3.19), approximating a normal distribution.

The Jarque-Bera test results presented in Table 1 demonstrated that all variables significantly rejected the normality hypothesis at the 1 % significance level, whilst the ERS unit root test confirmed that all series exhibited stationarity, thereby satisfying the fundamental prerequisite

conditions for the subsequent QQKRLS and QQGC analyses.

Table 2 presents the results of Pearson correlation tests, which reveal the linear correlation characteristics between the GBI and various national stock markets, alongside inter-market correlations amongst the stock markets themselves. The GBI demonstrates the strongest correlation with the US (0.833), whilst maintaining robust positive correlations with CAN (0.603), GER (0.577), and FRA (0.56). Conversely, the correlation between the GBI and the CHN was the weakest (0.247).

The dynamic time-series trends illustrated in Fig. 1 demonstrate the evolutionary trajectories and volatility characteristics of the GBI and stock markets across the G7 countries and China throughout the period from 2020 to 2025. From a temporal perspective, all series sustained severe negative shocks in early 2020, which corresponded closely with the systemic risk propagation across global financial markets during the initial outbreak of the COVID-19 pandemic. The GER demonstrated the most significant decline (approaching −20 %), whilst the CHN exhibited relatively moderate volatility.

Following mid-2020, stock markets across all countries gradually returned to relatively stable volatility ranges, with most series fluctuating within a ± 5 % range, demonstrating the markets' self-correcting capacity and the effectiveness of policy interventions. It was noteworthy that the GBI displayed relatively stable volatility characteristics

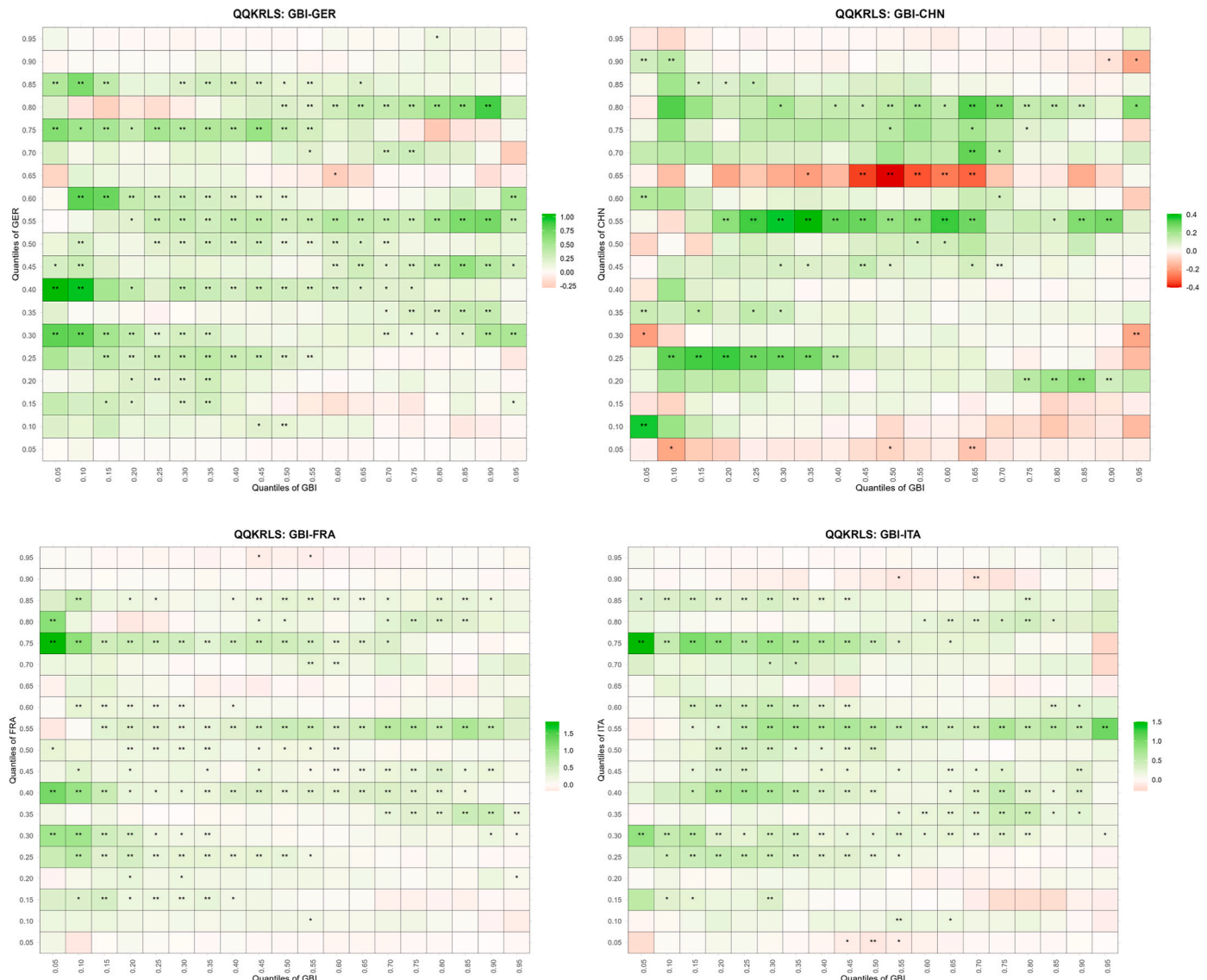


Fig. 2. Heatmap of the QQKRLS of GBI on G7 and Chinese stock markets.

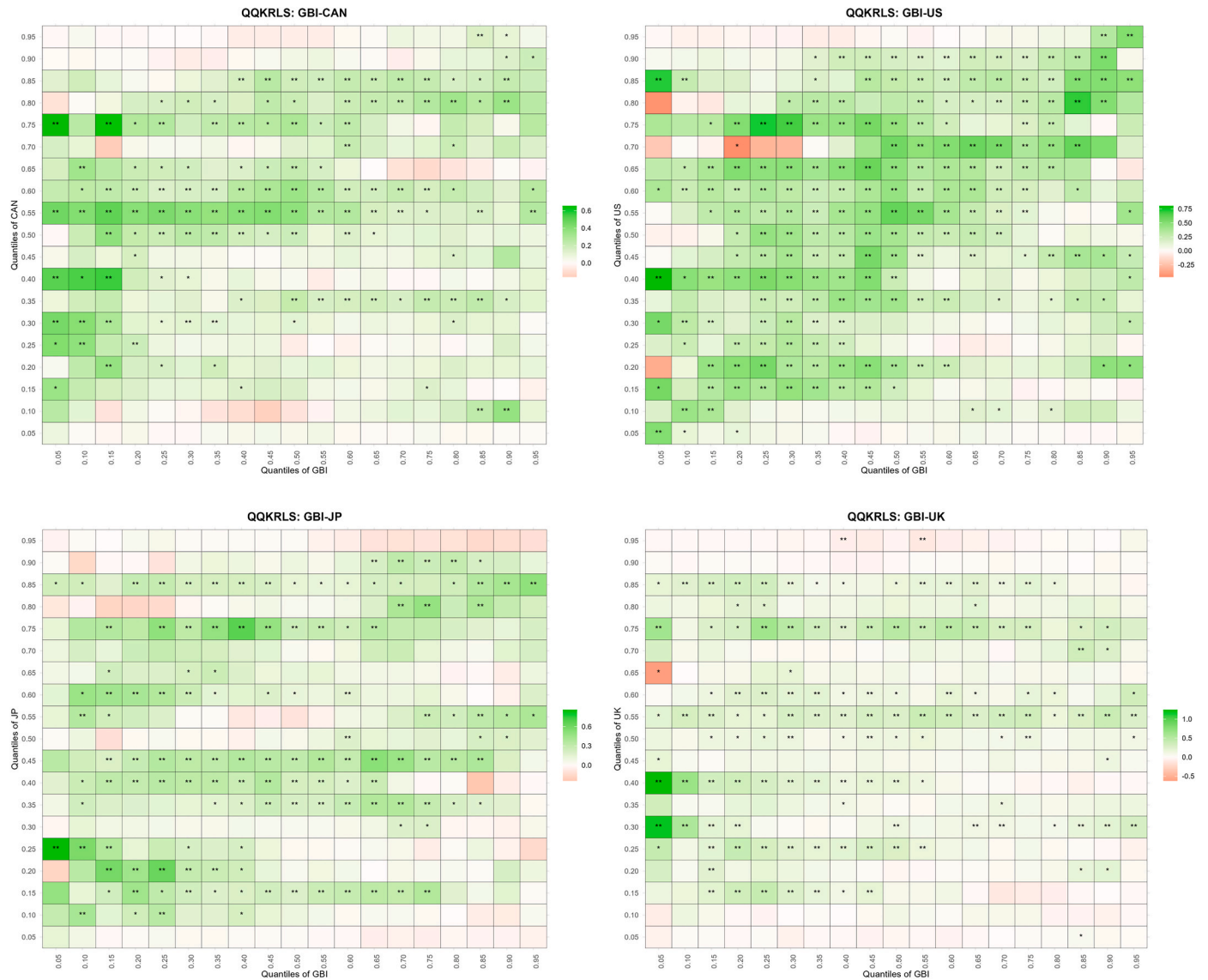


Fig. 2. (continued).

throughout the entire observation period, with comparatively fewer extreme value events, which aligned with the inherent characteristics of biodiversity as a long-term environmental indicator.

The synchronisation amongst national stock markets proved particularly evident at specific temporal junctures, most notably during the pandemic shock period in 2020 and several critical nodes between 2021 and 2022. These patterns reflect the risk contagion effects within financial markets operating under globalisation. The CHN exhibited relatively independent positive volatility in late 2024, with diminished synchronisation compared to other markets, which researchers could potentially attribute to the distinctive characteristics of China's economic policy cycles and market structure (Shao et al., 2024).

Based on the empirical analysis results presented in Fig. 2, which were derived from the QQKRLS methodology, this study employed the QQKRLS approach to conduct an in-depth examination of the heterogeneous effects of the GBI on the stock markets of the G7 countries and China. This methodology effectively captured the nonlinear relationships and parameter instability between variables, thereby providing a more precise analytical framework for understanding the transmission mechanisms of biodiversity indices across different market environments. In the heat maps of QQKRLS estimation results, green areas represented positive effect coefficients, red areas indicated negative

effect coefficients, whilst colour intensity reflected the magnitude of effects. Statistical significance was denoted through *** (1 % level), ** (5 % level), and * (10 % level).

Concerning the impact of GBI on the GER, the research identified a significant positive correlation between the two variables, with this relationship remaining stable across most quantile intervals. The graphical representation revealed intense green colouring in the convergence area between GBI's lower quantiles (0.05–0.25) and the GER's 0.75–0.85 quantiles, indicating that even when the biodiversity index demonstrated relatively weak performance, the GER's robust performance could still maintain positive associations. More significantly, within the extensive green area formed by GBI's middle-to-high quantiles (0.40–0.95) and the GER's middle-to-high quantiles (0.40–0.95), positive effects were particularly pronounced and almost entirely statistically significant, with coefficient values generally fluctuating within the 0.25–1.00 range. Particularly noteworthy was the formation of a deep green high-coefficient area around GBI's 0.40 quantile and the GER's 0.40 quantile, demonstrating the strongest positive influence. This phenomenon potentially reflected Germany's position as a European economic powerhouse, with its capital market exhibiting heightened sensitivity and positive responsiveness to sustainable development and biodiversity protection issues.

The Chinese stock market's response pattern to GBI presented more complex and distinctly layered characteristics. The red area in the lower-left corner of the graph clearly demonstrated significant negative impacts at the convergence of GBI's 0.05–0.15 quantiles and the Chinese stock market's 0.05–0.35 quantiles, with coefficient values ranging approximately between -0.2 and -0.4 , achieving statistical significance at the 0.05 and 0.30 quantiles. This negative relationship re-emerged around GBI's 0.45–0.70 quantiles and the Chinese stock market's 0.65 quantile vicinity, forming a distinct red band-like area. However, the central region of the graph presented a markedly different scenario: within the extensive green area formed by GBI's 0.25–0.85 quantiles and the Chinese stock market's 0.55–0.85 quantiles, positive relationships dominated, with coefficient values generally ranging between 0.1 and 0.4 and most achieving statistical significance. Particularly notable was the deep green stripe formed around the 0.55 quantile, indicating the strongest positive effects. This transformation pattern from negative to positive could potentially stem from phased differences in the Chinese capital market's recognition and acceptance of biodiversity concepts, whilst also reflecting the gradual characteristics of China's green finance development process.

The French stock market demonstrated extremely stable and consistent positive response characteristics to GBI. The entire heat map was almost dominated by green colouring, with only neutral or weakly negative pale areas appearing in very limited regions. Across GBI's entire quantile range of 0.05–0.95, a vast continuous green area formed

with the French stock market's 0.25–0.90 quantiles, with coefficient values stably maintained between 0.5 and 1.5, displaying persistent and robust positive influences. Particularly striking was the formation of a deep green peak area with coefficient values approaching 1.5 at the convergence of GBI's 0.75 quantile and the French stock market's 0.75 quantile, possessing high statistical significance. Even in lower quantile intervals, such as the area between GBI's 0.30–0.50 quantiles and the French stock market's 0.30–0.60 quantiles, positive relationships remained robust and statistically significant. As the originator of the Paris Agreement and an active promoter of global climate change governance, France's capital market possessed natural affinity towards environmental issues such as biodiversity.

The Italian stock market's response to GBI exhibited pronounced quantile dependency and asymmetric distribution characteristics. The lower-left corner area of the graph indicated that at the convergence of GBI's 0.05–0.25 quantiles and the Italian stock market's 0.05–0.30 quantiles, relationships were relatively weak and predominantly characterised by pale areas, lacking statistical significance. However, as quantile levels increased, green areas gradually expanded and colours deepened. Within the diagonal green band area formed by GBI's 0.30–0.85 quantiles and the Italian stock market's 0.30–0.85 quantiles, positive relationships became increasingly evident, with coefficient values fluctuating between 0.5 and 1.5 and most achieving statistical significance. Particularly noteworthy was the formation of a deep green core area in the middle quantile intervals (0.40–0.60), demonstrating

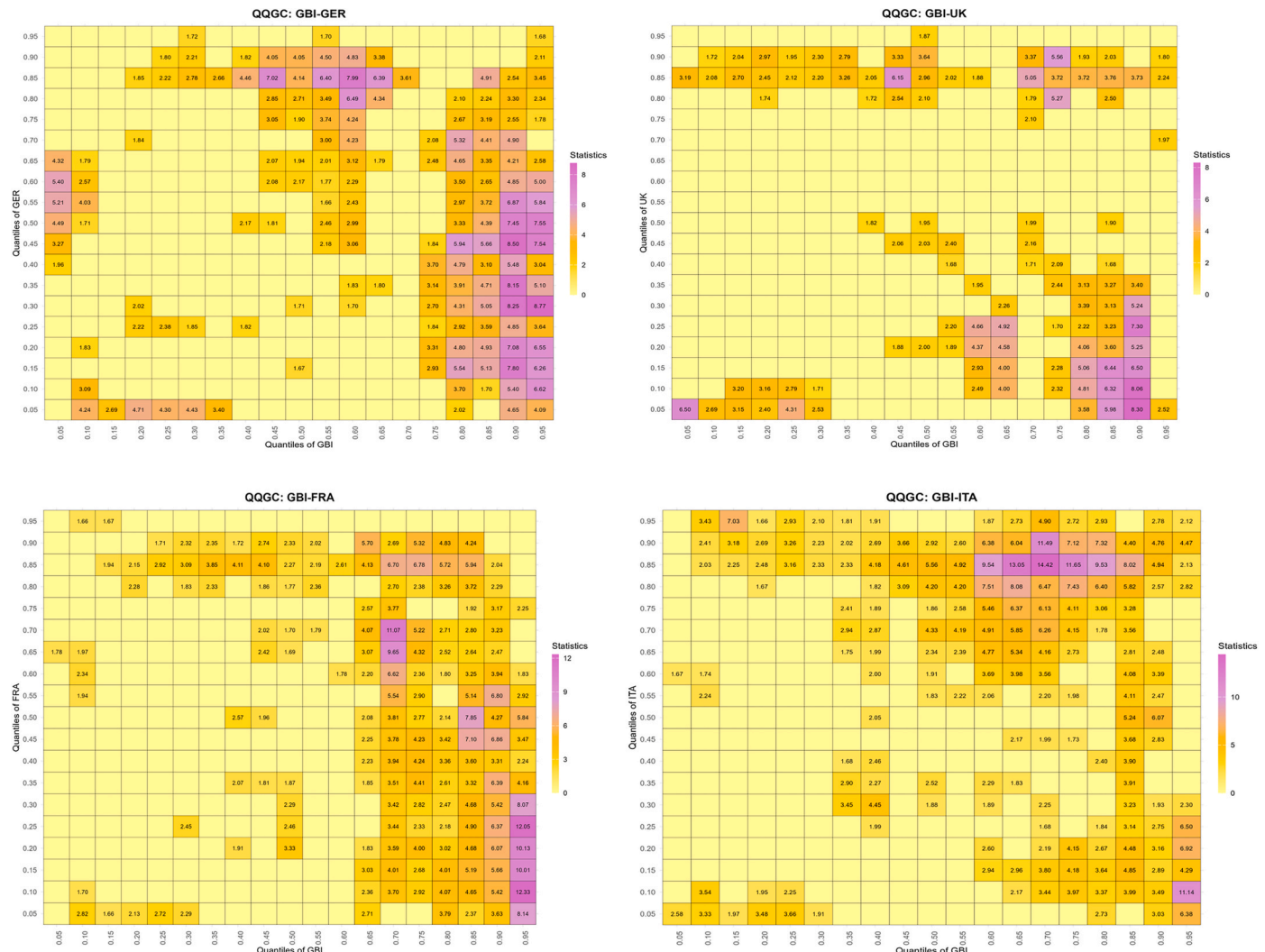


Fig. 3. QQGC of GBI for G7 and Chinese stock markets.

the strongest positive effects. The upper-right corner's 0.85–0.95 quantile area, although displaying relatively lower coefficient values, still maintained positive relationships.

The Canadian stock market demonstrated overall positive responses to GBI but presented a distinctive banded distribution pattern. The graph revealed that green areas were primarily concentrated in two key regions: firstly, the upper green band formed by GBI's 0.05–0.20 quantiles and the Canadian stock market's 0.75–0.95 quantiles, with coefficient values approximately 0.2–0.6, indicating that even when biodiversity index performance was relatively weak, strong Canadian stock market performance could still generate positive linkages. Secondly, the main green area formed by GBI's 0.40–0.95 quantiles and the Canadian stock market's 0.35–0.95 quantiles presented an extensive area with coefficient values stably ranging between 0.2 and 0.6, almost entirely achieving statistical significance. The intermediate 0.20–0.40 quantile area presented relatively neutral pale colouring, indicating weaker influence intensity. As a country rich in natural resources, Canada's balance between economic development and ecological environmental protection had consistently attracted considerable attention.

The US stock market's response mechanism to GBI demonstrated high complexity and refined quantile sensitivity. The mixed-colour area in the lower-left corner of the graph revealed complex patterns of positive and negative alternation in lower quantile intervals: the area between GBI's 0.05–0.25 quantiles and the US stock market's 0.05–0.25 quantiles exhibited red patches indicating negative relationships, whilst adjacent areas displayed positive relationships. As quantile levels increased, green areas gradually assumed dominance. Within the extensive green area formed by GBI's 0.30–0.95 quantiles and the US

stock market's 0.40–0.95 quantiles, positive relationships were robust with coefficient values fluctuating between 0.25 and 0.75, with most achieving statistical significance. Particularly in high quantile intervals (0.85–0.95), green colouring was most intense, demonstrating the strongest positive effects. The graph also revealed distinct diagonal patterns, indicating that correlations were most stable when both variables were at similar quantile levels.

The JP's response to GBI exhibited distinctive block-like distribution and phased characteristics. The graphical representation revealed pronounced segmented features: within GBI's 0.05–0.30 quantile intervals, responses from the Japanese stock market presented a mosaic pattern of positive and negative alternation, with a significant deep green band forming around the 0.25 quantile, where coefficient values approached 0.6 and possessed high statistical significance. In the middle quantile intervals (0.35–0.70), green areas were relatively continuous and stable, with coefficient values maintained between 0.3 and 0.6, demonstrating stable positive relationships. However, in high quantile intervals (0.75–0.95), the graph presented more complex patterns, with green areas and pale areas appearing alternately, indicating that relationship strength had somewhat diminished whilst remaining predominantly positive. This phenomenon could potentially be closely related to the emphasis on environmental responsibility within Japanese corporate culture and the Japanese government's international commitments regarding biodiversity protection.

The UK stock market demonstrated relatively stable and extensive positive response patterns to GBI. The entire heat map was dominated by green tones, displaying persistent positive relationships. Across GBI's entire range of 0.05–0.95, a continuous green area was formed with the

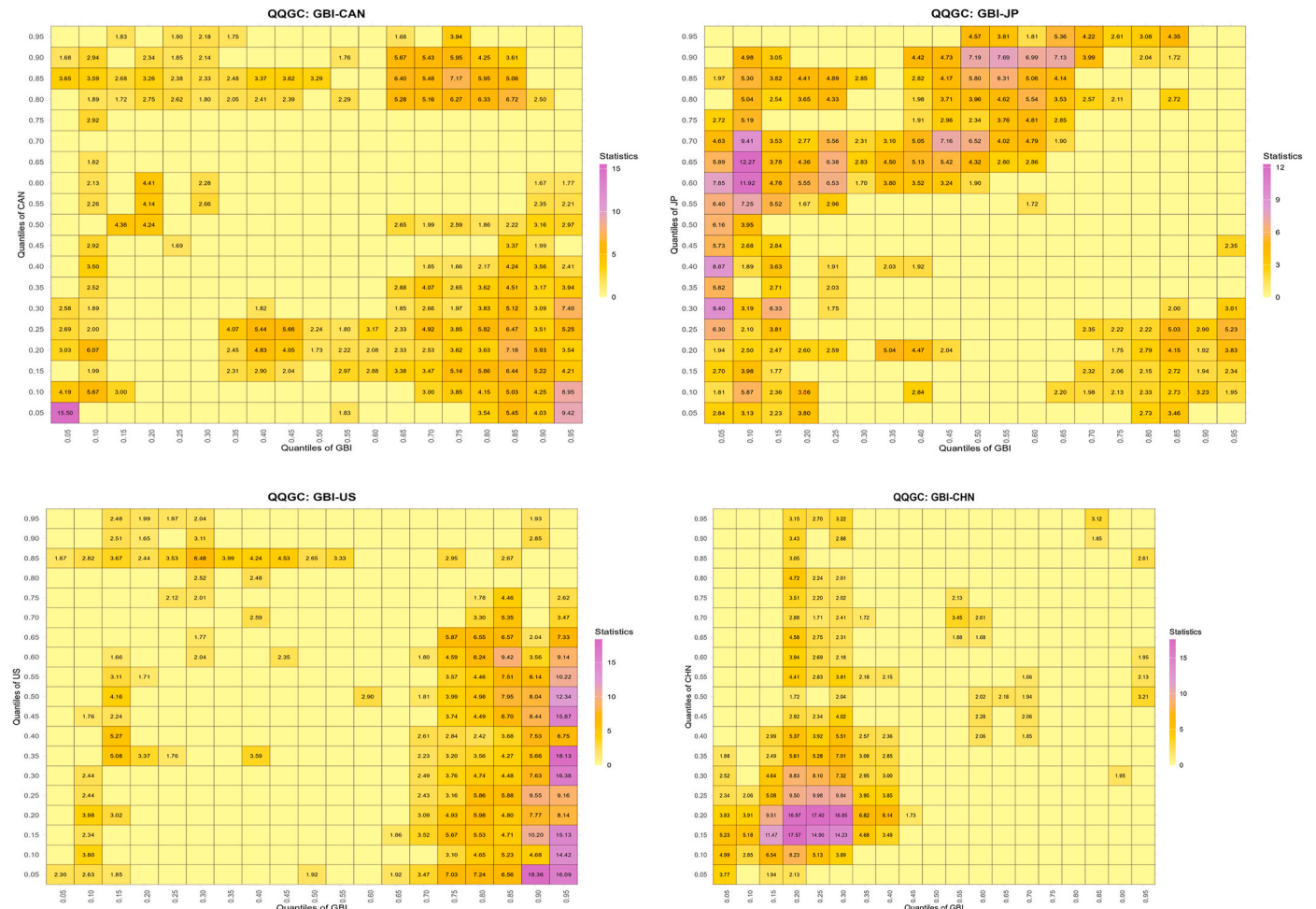


Fig. 3. (continued).

UK stock market's 0.30–0.85 quantiles, with coefficient values stably maintained between 0.5 and 1.0 and most possessing statistical significance. Particularly noteworthy was the formation of a deep green core area in the middle quantile intervals (0.40–0.60), demonstrating the strongest positive effects. Although certain areas in the upper-right and lower-left corners of the graph displayed lighter colours, they still maintained positive relationships. A small-scale red area appeared around the 0.65 quantile, but this negative influence was relatively limited and lacked statistical significance.

Through horizontal comparative analysis of the heat maps across countries, several important common characteristics and differential patterns could be identified. Firstly, all developed countries except China exhibited a fundamental pattern dominated by positive relationships, indicating that biodiversity indices exerted generally positive influences on developed country stock markets. Secondly, most countries presented quantile dependency, whereby influence intensity and significance demonstrated obvious differences under different quantile combinations. Thirdly, positive relationships in high quantile intervals were generally more stable and significant, potentially reflecting that investors were more willing to focus on long-term sustainable development factors when market performance was favourable.

Comprehensive analysis revealed that GBI's influence on G7 countries and Chinese stock markets exhibited significant heterogeneous characteristics. This heterogeneity manifested itself not only in differences between various countries but also in the variation patterns at different quantile levels within the same country. Stock markets in developed countries generally demonstrated positive responses to biodiversity indices, although response intensity and stability varied, primarily stemming from differences across countries in environmental policies, market maturity, investor structures, and cultural backgrounds. The application of the QKRLS methodology revealed complex relationship patterns that traditional linear models could not capture, providing important empirical evidence for understanding the transmission mechanisms of sustainable finance within global capital markets. Simultaneously, these findings offered valuable references for investors in formulating quantile-sensitive investment strategies and for policymakers in refining green finance regulatory frameworks.

In Fig. 3, the QQGC methodology was employed to analyse comprehensively the predictive capacity of the Standard & Poor's Global Large and Mid Cap Biodiversity Index (GBI) on stock markets of the G7 countries and China. The advantage of the QQGC method lies in its simultaneous consideration of the conditional distributions of both dependent and independent variables, enabling the capture of heterogeneous impacts of biodiversity indices on various national stock markets under different market conditions, thereby providing a more refined analytical framework for understanding the transmission mechanisms of ESG factors within global financial markets.

For Germany, QQGC results demonstrated that GBI exhibited significant predictive capacity across most quantile combinations. Particularly noteworthy was the observation that when GBI was positioned at middle-to-low quantiles (0.15–0.45), it generated strong causal relationships with Germany's high quantiles (0.80–0.95), with statistical values exceeding 7.0 and presenting deep red regions. This pattern indicated that deterioration in biodiversity conditions could potentially provide significant early warning effects for extreme upward movements in the German market. Simultaneously, when both GBI and Germany were positioned within middle quantile ranges, causal relationships remained relatively moderate yet stable, reflecting the persistent influence of biodiversity risks on the normalised performance of the German market.

The UK stock market's QQGC spectrum presented relatively dispersed causal relationship patterns. GBI's predictive capacity for the UK was primarily concentrated in specific quantile intersection regions, particularly when GBI was positioned at extremely low quantiles (0.05–0.15) and extremely high quantiles (0.85–0.95), demonstrating relatively strong predictive power for the UK's middle-to-high quantile

performance. This discontinuous causal relationship pattern potentially reflected that the UK's sensitivity to global ESG risks, as a financial centre, exhibited phased characteristics, with market reactions becoming more acute under extreme states of biodiversity risk.

France demonstrated the most intensive and continuous causal relationship network. From the heat map, it could be observed that GBI's predictive capacity for France exhibited statistical significance across almost all quantile combinations, particularly in the region where GBI's middle-to-low quantiles (0.25–0.65) intersected with France's high quantiles (0.70–0.95), where statistical values generally exceeded 5.0, forming extensive orange-red regions. This finding indicated that the French stock market possessed high sensitivity to biodiversity risks, with changes in biodiversity indices being able to comprehensively and stably predict French stock market performance across different market states, reflecting France's high degree of internalisation of environmental risks under the EU's ESG regulatory framework.

The Italian causal relationship patterns presented pronounced quantile-dependent characteristics. GBI's predictive capacity for Italy was primarily manifested in two key regions: firstly, GBI's high quantiles (0.70–0.95) generated strong causal relationships with Italy's high quantiles (0.80–0.95), with statistics exceeding 10.0; secondly, GBI's middle quantiles (0.35–0.65) formed stable predictive capacity with Italy's middle quantiles (0.45–0.75). This bimodal distribution of causal relationship structure suggested that the Italian stock market's response mechanisms to biodiversity risks exhibited significant differences across various market states.

Canada exhibited relatively moderate yet broadly distributed causal relationship characteristics. GBI's predictive capacity for Canada maintained moderate-intensity statistical significance across multiple quantile intervals, particularly in regions where GBI's low quantiles (0.05–0.25) intersected with Canada's high quantiles (0.80–0.95), and where GBI's high quantiles (0.75–0.95) intersected with Canada's middle quantiles (0.35–0.65). This relatively balanced distribution of causal relationships could potentially be related to Canada's abundant natural resource endowments and relatively mature environmental regulatory systems, enabling its stock market to respond to biodiversity risks in a more stable and predictable manner.

Japan presented distinctive causal relationship patterns, with the primary characteristic being the formation of high-intensity causal relationship clusters at specific quantile intersection points. The most significant region appeared within the range where GBI's middle-to-low quantiles (0.25–0.45) intersected with Japan's middle-to-low quantiles (0.25–0.55), with statistical values exceeding 12.0, forming deep purple high-intensity regions. This concentrated causal relationship pattern indicated that the Japanese stock market's response to biodiversity risks possessed obvious threshold effects, with market reactions becoming exceptionally sensitive at specific risk levels.

The US stock market's QQGC results revealed relatively complex causal relationship structures. GBI's predictive capacity for the US was primarily concentrated in two key regions: when GBI was positioned at middle quantiles (0.40–0.70), it generated significant causal relationships with US high quantiles (0.75–0.95); when GBI was positioned at high quantiles (0.80–0.95), it formed strong predictive power with US middle-to-low quantiles (0.25–0.55), with statistics exceeding 15.0. This cross-sectional causal relationship pattern reflected the complex response mechanisms of the United States, as the world's largest capital market, to biodiversity risks.

The Chinese causal relationship patterns presented distinctive asymmetric characteristics. GBI's predictive capacity for China was primarily concentrated in regions where GBI's low quantiles (0.10–0.40) intersected with China's low quantiles (0.10–0.45), and where GBI's extremely low quantiles (0.05–0.20) intersected with China's middle quantiles (0.25–0.65). Notably, when GBI was positioned at extremely low quantiles (0.15–0.25), it generated exceedingly high statistics exceeding 17.0 for China's low quantiles (0.15–0.25), forming the deepest purple regions. This finding indicated that the Chinese stock

Table 3

Quantile regression of GBI on G7 and Chinese stock markets.

Index	Q10	Q25	Q50	Q75	Q90
GER	0.6907	0.2485	0.3002	0.3329	−0.0281
UK	0.748	0.3313	0.1698	0.3075	0.2297
ITA	0.6215	0.228	0.2969	0.2106	−0.1326
CAN	0.9266	0.23	0.2838	0.2449	0.3897
JP	0.6666	0.1316	0.0871	0.2899	0.0621
US	0.932	0.3207	0.6298	0.3668	0.7596
CHN	0.4248	0.1037	−0.0528	0.1564	0.0191
FRA	0.6495	0.2102	0.2739	0.3341	0.0309

Notes: We used five interquartile levels, 10 %, 25 %, 50 %, 75 %, and 90 % as the quantiles chosen for the work.

market demonstrated significant vulnerability during periods of biodiversity risk deterioration, with sensitivity to environmental risks being particularly pronounced during market downturns.

Comprehensively, QQGC analysis revealed significant heterogeneity and complexity in GBI's influence on different national stock markets. France and Germany demonstrated comprehensive sensitivity to biodiversity risks, reflecting the EU's stringent ESG regulatory environment; US and Japanese stock markets presented phased and threshold-based response characteristics, embodying the complex pricing mechanisms of developed markets towards environmental risks; whilst the Chinese stock market displayed obvious vulnerability during periods of biodiversity risk deterioration, suggesting that emerging markets still possess room for improvement in environmental risk management. These findings provide important empirical evidence for investors in formulating biodiversity risk-based investment strategies under different market conditions, whilst simultaneously offering valuable references for policymakers in understanding the transmission mechanisms of environmental risks within the global financial system.

3.2. Robustness test

Table 3 presents robustness testing results based on quantile regression methodology, which provide strong validation for the core findings from the QKRLS and QQGC primary analyses. The table displays regression coefficients showing GBI's effects on various national stock markets across five critical quantile levels (0.10, 0.25, 0.50, 0.75, 0.90). The results demonstrate high consistency with the primary analytical findings mentioned above.

In Table 3, the German stock market's coefficients exhibited a progressive decline from 0.6907 at the 0.10 quantile to −0.0281 at the 0.90 quantile, whilst the French stock market demonstrated a corresponding decrease from 0.6495 to 0.0309. Both markets displayed pronounced quantile dependency that completely aligned with the causal relationship intensity variation patterns observed in the QQGC analysis. The Chinese stock market's coefficients revealed a distinctive U-shaped trajectory, transitioning from 0.4248 to negative values of −0.0528 before recovering to 0.0191, precisely corroborating the complex nonlinear relationship structures identified in the QKRLS heat maps.

The United States (0.932–0.7596) and Canada (0.966–0.397) maintained robust positive coefficients across all quantiles, entirely consistent with the stable positive causal relationships identified in the primary analysis. The Italian and Japanese stock markets similarly demonstrated declining patterns from strong positive to weak positive or negative coefficients, further confirming the robustness of the primary analytical results.

This cross-methodological consistency not only enhanced the

reliability of the research conclusions but, more importantly, confirmed that the quantile heterogeneity of biodiversity index influences on different national stock markets constituted a robust empirical regularity. The convergent findings across multiple analytical frameworks strengthened the validity of the observed patterns and established the quantile-dependent nature of these relationships as a fundamental characteristic of the biodiversity-financial market nexus.

The further robustness test results based on ordinary least squares (OLS) methodology, presented in Table 4, demonstrated significant positive correlations between GBI and all national stock markets, with coefficients ranging from 0.3497 to 0.9676. The United States exhibited the strongest correlation with GBI (0.9676), followed by Canada (0.8715) and Germany (0.7991), whilst China displayed a relatively weaker correlation (0.3497). These findings further validated the fundamental directional results of the QKRLS and QQGC primary analyses from a linear relationship perspective, confirming that biodiversity indices exert positive influences on most national stock markets.

The observation proves particularly noteworthy that although the OLS methodology failed to capture the complex quantile heterogeneity and nonlinear characteristics that QKRLS and QQGC revealed, the correlation intensity rankings it identified remained fundamentally consistent with the overall patterns of response intensity across countries in the primary analysis. The relatively weaker correlation of the Chinese stock market also aligned with the complex causal relationship patterns that the primary analysis identified, further corroborating the distinctive nature of emerging markets' response mechanisms to biodiversity risks.

4. Conclusions and policy remarks

This paper used advanced econometric methodologies, including quantile-on-quantile kernel-based regularised least squares (QKRLS) and quantile-on-quantile Granger causality (QQGC), to examine comprehensively the heterogeneous impact mechanisms of the Standard & Poor's Global Large and Mid Cap Biodiversity Index (GBI) on stock markets across G7 countries and China. The empirical results revealed complex transmission patterns of biodiversity risks within global financial markets, providing crucial empirical evidence for understanding the cross-national differential effects of sustainable finance. The research findings demonstrated that GBI exerted influence on different national stock markets through significant quantile dependency and nonlinear characteristics, with this heterogeneity manifesting not only across national boundaries but also in response variations within identical markets under differing conditions.

From a cross-national comparative perspective, stock markets in European developed countries such as France, Germany, and the United Kingdom demonstrated highly sensitive positive responses to biodiversity indices, a phenomenon closely associated with the EU's stringent ESG regulatory framework and mature green finance systems. Although the US stock market similarly exhibited positive relationships, its response patterns proved more complex, demonstrating cross-sectional causal relationship structures under different quantile combinations, thereby reflecting the multi-layered pricing mechanisms through which the US capital market approaches environmental risks. Canadian and Japanese stock markets exhibited relatively stable yet varied positive response characteristics, embodying the different adaptation strategies through which resource-based economies and technology-oriented economies address biodiversity risks. The Chinese stock market's response pattern proved most distinctive, presenting pronounced asymmetric and phased characteristics that demonstrated significant

Table 4

OLS regression.

Index	GER	UK	ITA	CAN	JP	US	CHN	FRA
Correlation	0.7991	0.7338	0.7276	0.8715	0.7363	0.9676	0.3497	0.7733

vulnerability during periods of biodiversity risk deterioration whilst exhibiting positive market reactions during periods of risk improvement. This complex pattern reflected the gradual characteristics of China's capital market during the green transformation process and the distinctiveness of its institutional environment.

Based on the empirical findings of this study, the following recommendations are proposed to policymakers across countries. Firstly, governments should strengthen policy coordination between biodiversity protection and financial market regulation, establishing systematic green finance risk monitoring and assessment systems. European countries should continue to assume leading roles in ESG regulation, further improving biodiversity-related information disclosure standards and risk assessment frameworks, thereby providing reference templates for the formulation of global green finance standards. The United States, as the world's largest capital market, should accelerate the advancement of federal-level climate and environmental risk regulatory legislation, unifying regulatory standards across states and improving the consistency and effectiveness through which markets price environmental risks.

For emerging market countries, particularly China, policy priorities should focus on gradually improving green finance infrastructure development. Establishing more refined biodiversity risk classification and rating systems is recommended, alongside strengthening mandatory requirements for listed companies' environmental information disclosure, whilst simultaneously guiding capital allocation towards sustainable development sectors through policy instruments such as tax incentives and green bonds. Given the complex response patterns that Chinese stock markets exhibit towards biodiversity risks, regulatory authorities should establish more flexible classified regulatory mechanisms, formulating differentiated environmental risk management requirements for different industries and enterprise types. Additionally, strengthening international cooperation remains crucial, with countries needing to enhance coordination of financial regulatory policies under international frameworks for biodiversity protection, avoiding regulatory arbitrage behaviours and promoting effective allocation of global green capital.

Central banks and financial regulatory authorities should incorporate biodiversity risks into macroprudential regulatory frameworks, establishing identification, monitoring, and response mechanisms for systemic environmental risks. Particularly when formulating monetary policies and financial stability policies, potential impacts of environmental risks on financial systems should receive full consideration, with policy instruments and operational approaches adjusted accordingly. Simultaneously, financial institutions should receive encouragement to strengthen environmental risk management capacity building, improve internal control and risk assessment systems, and enhance the degree of greenness and sustainable development capabilities of financial systems.

Based on the complex relationship patterns between biodiversity indices and national stock markets that this study revealed, targeted recommendations are proposed to different types of investors. For institutional investors, biodiversity risks should receive incorporation as important considerations in investment decision-making, establishing systematic ESG investment frameworks and risk assessment systems. When conducting cross-national asset allocation, the heterogeneous response characteristics that different markets exhibit towards biodiversity risks should receive full consideration, adopting quantile-sensitive investment strategies. For European market investments, investors could utilise the high sensitivity to biodiversity indices by closely monitoring environmental policy changes and ecological risk events to adjust portfolio weights and risk exposures.

For US market investment strategies, particular attention should focus on quantile cross-effects, adopting differentiated risk management strategies when markets exist in different states. Given the complex response patterns that US stock markets exhibit towards biodiversity risks, dynamic hedging strategies are recommended, adjusting portfolio environmental risk exposures according to changes in market quantile

levels. For Chinese market investments, particular attention must focus on the amplification effects of biodiversity risks during market downturns, with recommendations to allocate defensive assets related to environmental improvement within portfolios whilst monitoring the impact that phased changes in China's green transformation policies exert on market sentiment.

Individual investors should enhance their cognitive levels regarding environmental risks, incorporating biodiversity and other long-term sustainable development factors into investment decision-making considerations. Participation indirectly in sustainable investment through ESG-themed funds, green bonds, and other products is recommended, whilst monitoring the environmental performance and biodiversity protection measures that invested enterprises implement. When conducting international diversified investments, understanding the sensitivity differences that various markets exhibit towards environmental risks proves essential, with reasonable asset weight allocation optimising risk-return characteristics.

For risk managers and quantitative investment teams, incorporating the quantile dependency characteristics that this study discovered into risk models and investment strategy designs is recommended. Through constructing quantile-based dynamic risk adjustment mechanisms, the transmission effects of environmental risks under different market conditions can receive better capture. Simultaneously, developing specialised biodiversity risk factors and integrating them into multi-factor investment models is recommended to improve the long-term sustainability and risk-adjusted returns of portfolios. Investors should also establish comprehensive environmental risk monitoring systems, regularly assessing portfolio environmental risk exposures and timely adjusting investment strategies to respond to potential systemic environmental risk shocks.

Future research could further deepen and expand the findings of this study in several directions. Firstly, with the continuous improvement of ESG data quality and the constant emergence of new environmental indicators, future research could construct more comprehensive environmental risk indicator systems, combining biodiversity risks with other environmental risk factors such as climate change, water resource scarcity, and soil degradation, exploring the interactive mechanisms of multi-dimensional environmental risks and their joint impacts on financial markets. Secondly, considering that this study primarily focused on developed countries and China, future research should extend the analytical scope to encompass more emerging markets and developing countries, particularly those hotspot regions facing significant challenges in biodiversity protection, to test the universality of research conclusions and identify region-specific transmission mechanisms.

Conflicts of interest

There are no conflicts of interest to declare.

References

- Adebayo, T. S., & Özkan, O. (2024). Evaluating the role of financial globalization and oil consumption on ecological quality: A new perspective from quantile-on-quantile granger causality. *Heliyon*, 10(2), Article e24636.
- Adebayo, T. S., Özkan, O., & Eweade, B. S. (2024). Do energy efficiency R&D investments and information and communication technologies promote environmental sustainability in Sweden? A quantile-on-quantile KRLS investigation. *Journal of Cleaner Production*, 440, Article 140832.
- Balcilar, M., Bekiros, S., & Gupta, R. (2017). The role of news-based uncertainty indices in predicting oil prices: A hybrid nonparametric quantile causality method. *Empirical Economics*, 53(3), 879–889.
- Bank of England. (2021). Supervisory statement SS3/19. <https://www.bankofengland.co.uk/prudential-regulation/publication/2019/enhancing-banks-and-insurers-approaches-to-managing-the-financial-risks-from-climate-change-ss>.
- Baste, I. A., & Watson, R. T. (2022). Tackling the climate, biodiversity and pollution emergencies by making peace with nature 50 years after the Stockholm conference. *Global Environmental Change*, 73, Article 102466.
- Baur, D. G. (2013). The structure and degree of dependence: A quantile regression approach. *Journal of Banking & Finance*, 37(3), 786–798.

- Baur, D., & Schulze, N. (2005). Coexceedances in financial markets—a quantile regression analysis of contagion. *Emerging Markets Review*, 6(1), 21–43.
- Becchetti, L., Cordella, M., & Morone, P. (2022). Measuring investments progress in ecological transition: The green Investment financial Tool (GIFT) approach. *Journal of Cleaner Production*, 357, Article 131915.
- Chen, S., Song, Y., & Gao, P. (2023). Environmental, social, and governance (ESG) performance and financial outcomes: Analyzing the impact of ESG on financial performance. *Journal of Environmental Management*, 345, Article 118829.
- Chuang, C. C., Kuan, C. M., & Lin, H. Y. (2009). Causality in quantiles and dynamic stock return–volume relations. *Journal of Banking & Finance*, 33(7), 1351–1360.
- Engle, R. F., & Granger, C. W. (1987). Co-integration and error correction: Representation, estimation, and testing. *Econometrica*, 55(2), 251–276.
- Farooqi, T. J. A., Irfan, M., Portela, R., Zhou, X., Shulin, P., & Ali, A. (2022). Global progress in climate change and biodiversity conservation research. *Global Ecology and Conservation*, 38, Article e02272.
- Flammer, C., Giroux, T., & Heal, G. M. (2025). Biodiversity finance. *Journal of Financial Economics*, 164, Article 103987.
- Garel, A., Romec, A., Sautner, Z., & Wagner, A. F. (2024). Do investors care about biodiversity? *Review of Finance*, 28(4), 1151–1186.
- Hamilton, J. D. (1989). A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2), 357–384.
- Hansen, B. E. (1999). Threshold effects in non-dynamic panels: Estimation, testing, and inference. *Journal of Econometrics*, 93(2), 345–368.
- Hutchinson, M. C., & Lucey, B. (2024). A bibliometric and systemic literature review of biodiversity finance. *Finance Research Letters*, 64, Article 105377.
- Johansen, S. (1991). Estimation and hypothesis testing of cointegration vectors in Gaussian vector autoregressive models. *Econometrica*, 59(6), 1551–1580.
- Karolyi, G. A., & Tobin-de la Puente, J. (2023). Biodiversity finance: A call for research into financing nature. *Financial Management*, 52(2), 231–251.
- Koenker, R. (2005). *Quantile regression*. Cambridge University Press.
- Koenker, R., & Bassett, J. G. (1978). Regression quantiles. *Econometrica*, 46(1), 33–50.
- Kräussl, R., Oladiran, T., & Stefanova, D. (2024). A review on ESG investing: Investors' expectations, beliefs and perceptions. *Journal of Economic Surveys*, 38(2), 476–502.
- Ma, S., Zeng, H., & Abedin, M. Z. (2025). The impact of the reforms in the Chinese equities exchange and quotations on innovation in cross-border e-commerce enterprises. *Asia Pacific Business Review*, 1–41.
- McElwee, P., Chioleu-Assouline, M., Clapp, J., Isenhour, C., Jackson, T., Kelemen, E., ... Santos, R. (2020). Ensuring a post-COVID economic agenda tackles global biodiversity loss. *One Earth*, 3(4), 448–461.
- Mies, M. (2025). Biodiversity finance: Sovereign biodiversity loss drivers and financial stability. *Journal of Industrial Ecology*, 29(3), 937–954.
- Narula, R., Rao, P., Kumar, S., & Paltrinieri, A. (2025). ESG investing & firm performance: Retrospections of past & reflections of future. *Corporate Social Responsibility and Environmental Management*, 32(1), 1096–1121.
- NGFS. (2023). **Nature-related financial risks framework**. https://www.ngfs.net/sites/default/files/medias/documents/ngfs_conceptual-framework-on-nature-related-risks.pdf.
- Porter, M. E., & Linde, C. V. D. (1995). Toward a new conception of the environment-competitiveness relationship. *The Journal of Economic Perspectives*, 9(4), 97–118.
- Rands, M. R., Adams, W. M., Bennun, L., Butchart, S. H., Clements, A., Coomes, D., ... Vira, B. (2010). Biodiversity conservation: Challenges beyond 2010. *Science*, 329(5997), 1298–1303.
- Shahbaz, M., Van Hoang, T. H., Mahalik, M. K., & Roubaud, D. (2017). Energy consumption, financial development and economic growth in India: New evidence from a nonlinear and asymmetric analysis. *Energy Economics*, 63, 199–212.
- Shao, H., Zhou, B., Wang, D., & An, Z. (2024). Navigating uncertainty: The micro-level dynamics of economic policy uncertainty and systemic financial risk in china's financial institutions. *Journal of the Knowledge Economy*, 1–31.
- Sim, N., & Zhou, H. (2015). Oil prices, US stock return, and the dependence between their quantiles. *Journal of Banking & Finance*, 55, 1–8.
- Stone, C. J. (1985). Additive regression and other nonparametric models. *Annals of Statistics*, 13(2), 689–705.
- Warren, R., Price, J., VanDerWal, J., Cornelius, S., & Sohl, H. (2018). The implications of the united nations paris agreement on climate change for globally significant biodiversity areas. *Climatic Change*, 147(3), 395–409.
- Xiao, Z. (2009). Quantile cointegrating regression. *Journal of Econometrics*, 150(2), 248–260.
- Zeng, H., Benkraiem, R., Abedin, M. Z., & Hajek, P. (2025). Transitioning to sustainability: Dynamic spillovers between sustainability indices and Chinese stock market. *European Financial Management*. <https://doi.org/10.1111/eufm.12560>. Early View.